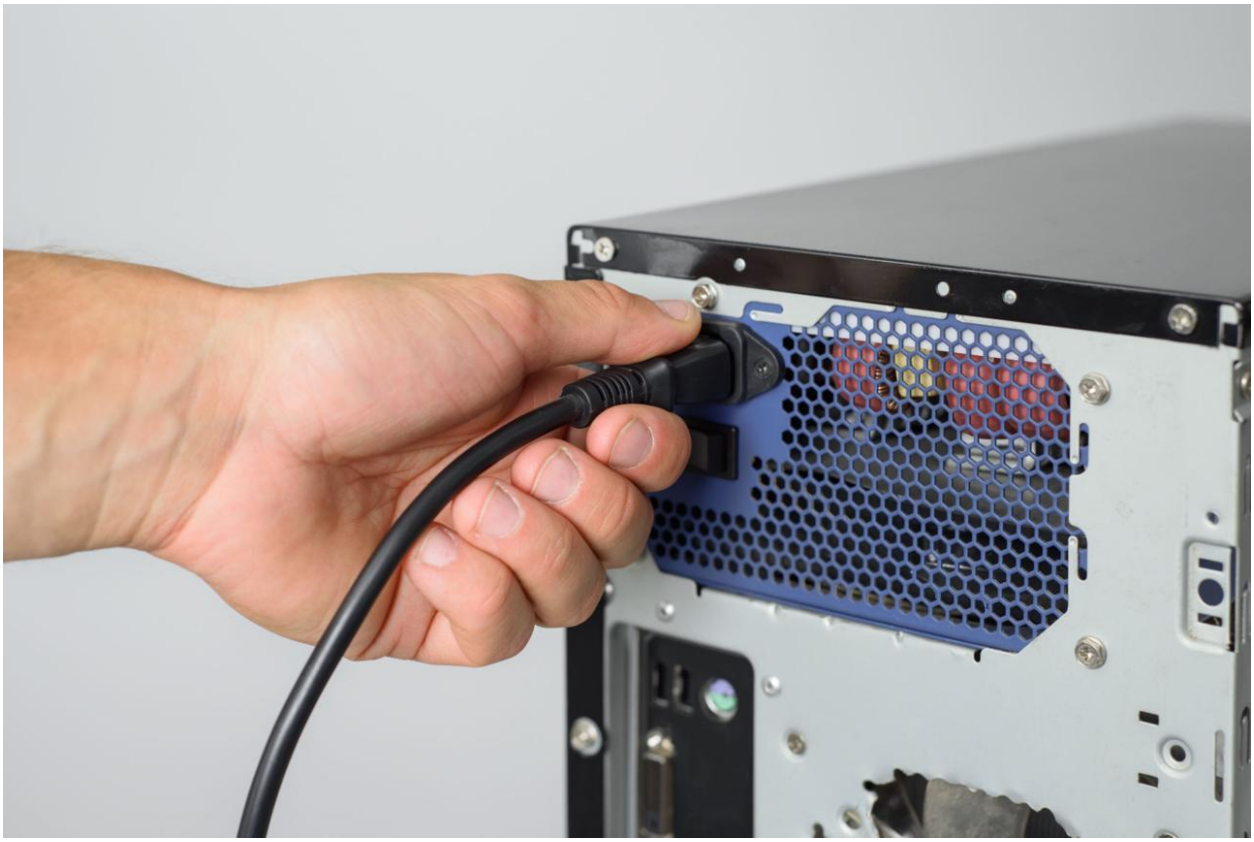
1. **List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.**

No.	Component	Description
1	Motherboard	The main circuit board that connects all computer components and allows them to communicate.
2	CPU (Processor)	The brain of the computer that processes instructions and performs calculations.
3	RAM	Temporarily stores data that the computer is currently using so programs can run smoothly.
4	HDD / SSD	Permanently stores the operating system, software, games, and personal files.
5	GPU (Graphics Card)	Processes graphics, images, and videos, especially for gaming and visual tasks.
6	PSU (Power Supply Unit)	Supplies electrical power to all the components inside the computer.
7	CPU Cooler	Keeps the processor cool and prevents it from overheating.
8	Cooling Fans	Circulate air inside the cabinet to maintain a suitable temperature.
9	Cabinet / Case	Holds and protects all the internal computer components.
10	CMOS Battery	Maintains the system clock and helps preserve BIOS/UEFI settings when the PC is switched off.
11	Expansion Cards	Add extra features such as Wi-Fi, Bluetooth, or improved sound capabilities.
12	Optical Drive	Reads or writes CDs and DVDs. It is optional and mostly found in older computers.
13	Power and Data Cables	Connect components and carry electrical power or data between them.

2. **Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**

Step 1: Switch off and unplug the PC



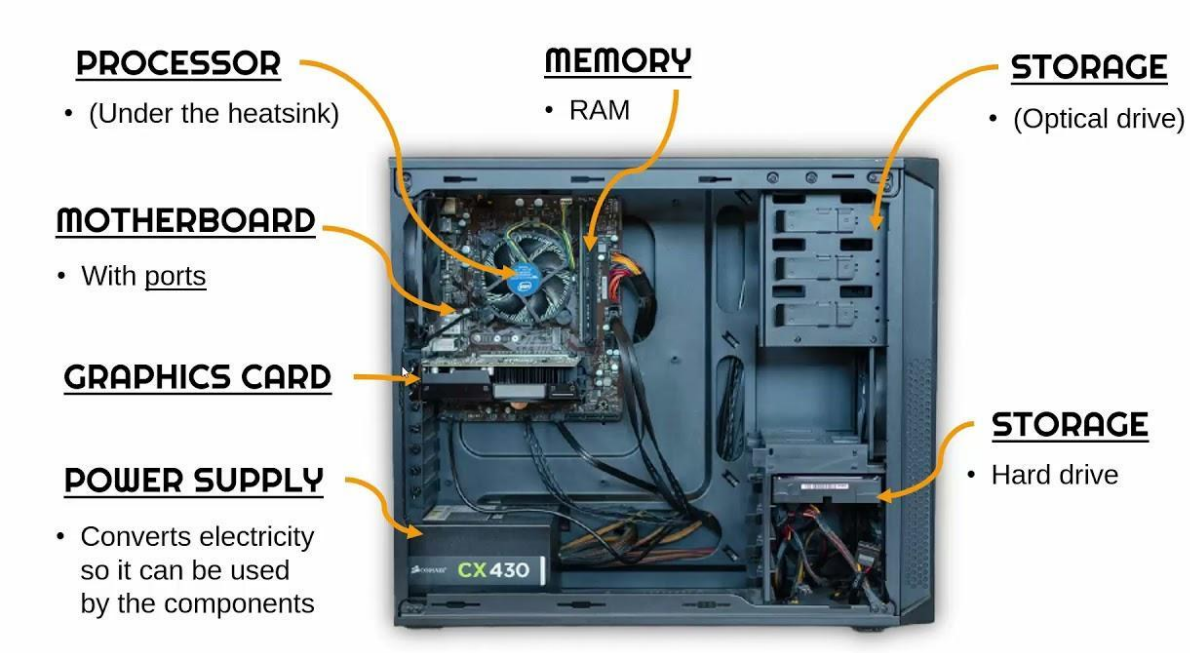
Caption: First, I switched off the computer and removed the power cable from the electrical socket. I also disconnected the keyboard, mouse, monitor, and other connected devices.

Step 2: Remove the cabinet side panel



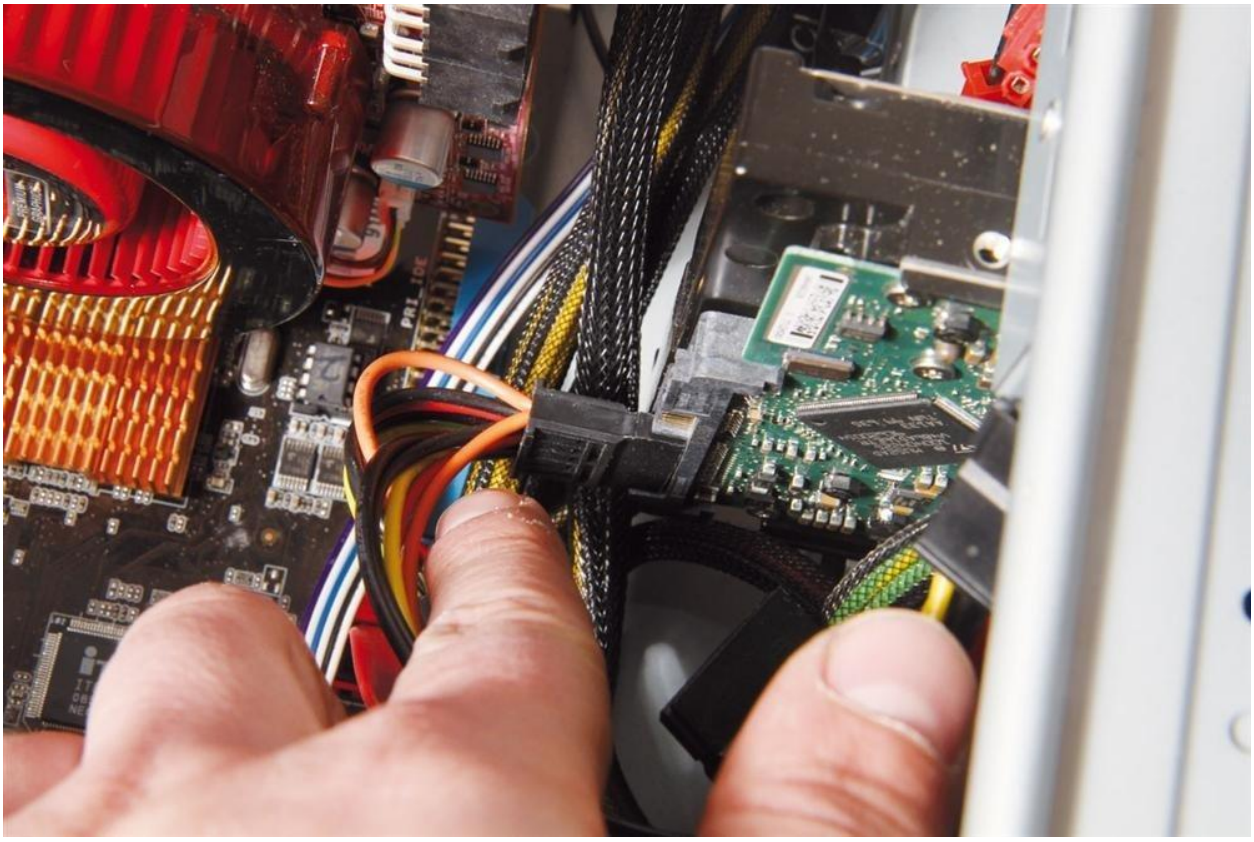
Caption: I opened the computer cabinet by removing the side panel screws. This allowed me to see all the internal components.

Step 3: Identify the internal components



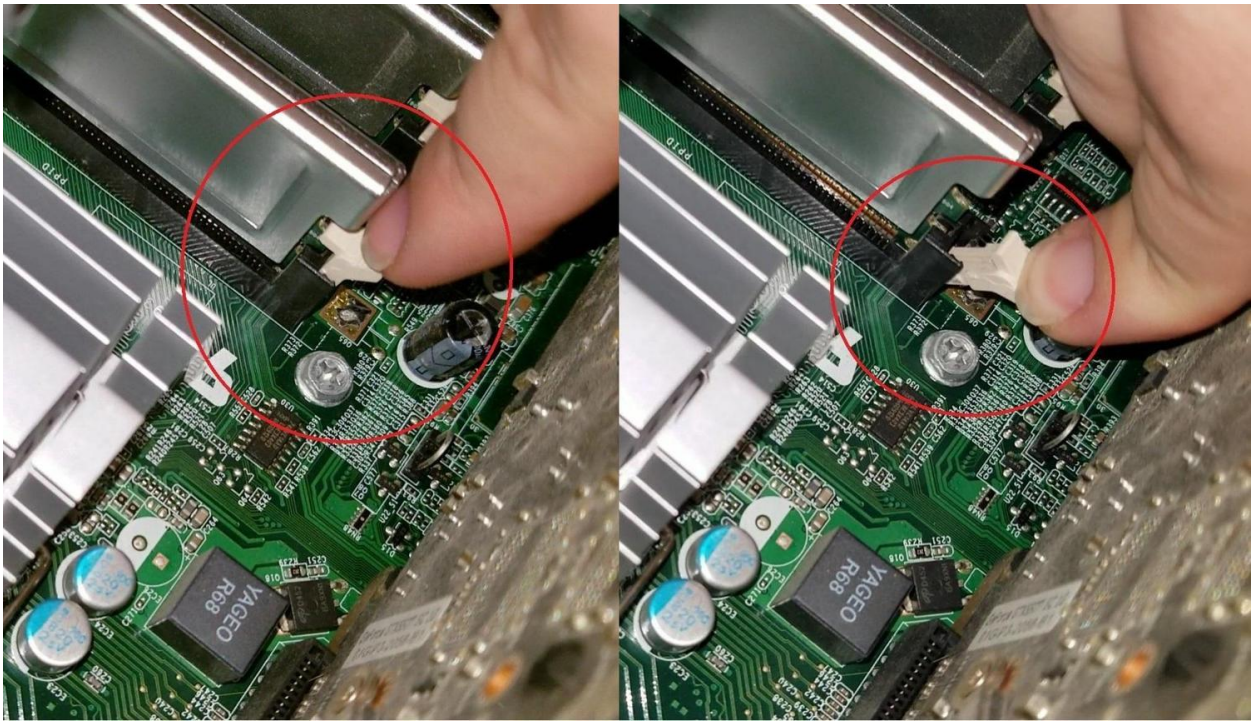
Caption: I identified the motherboard, CPU, RAM, storage drive, graphics card, power supply, and cooling fans inside the cabinet.

Step 4: Disconnect the cables



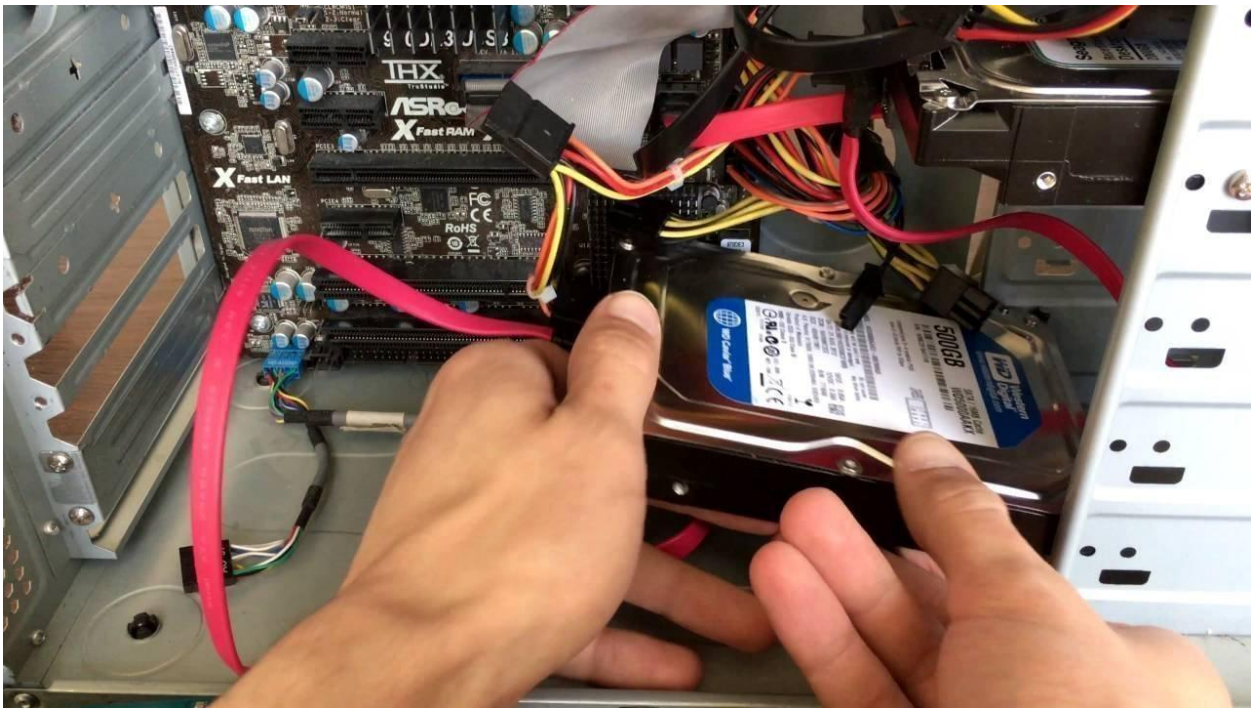
Caption: I carefully disconnected the power and data cables connected to the motherboard, storage drives, and other components.

Step 5: Remove the RAM



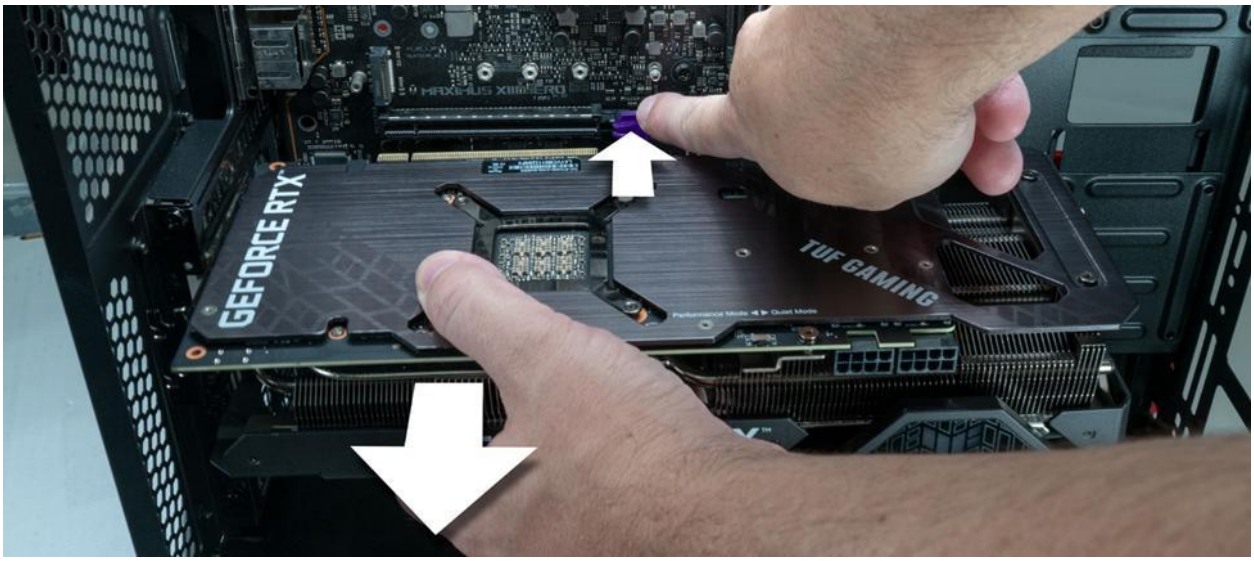
Caption: I opened the RAM slot clips and carefully removed the RAM sticks by holding their edges.

Step 6: Remove the HDD/SSD



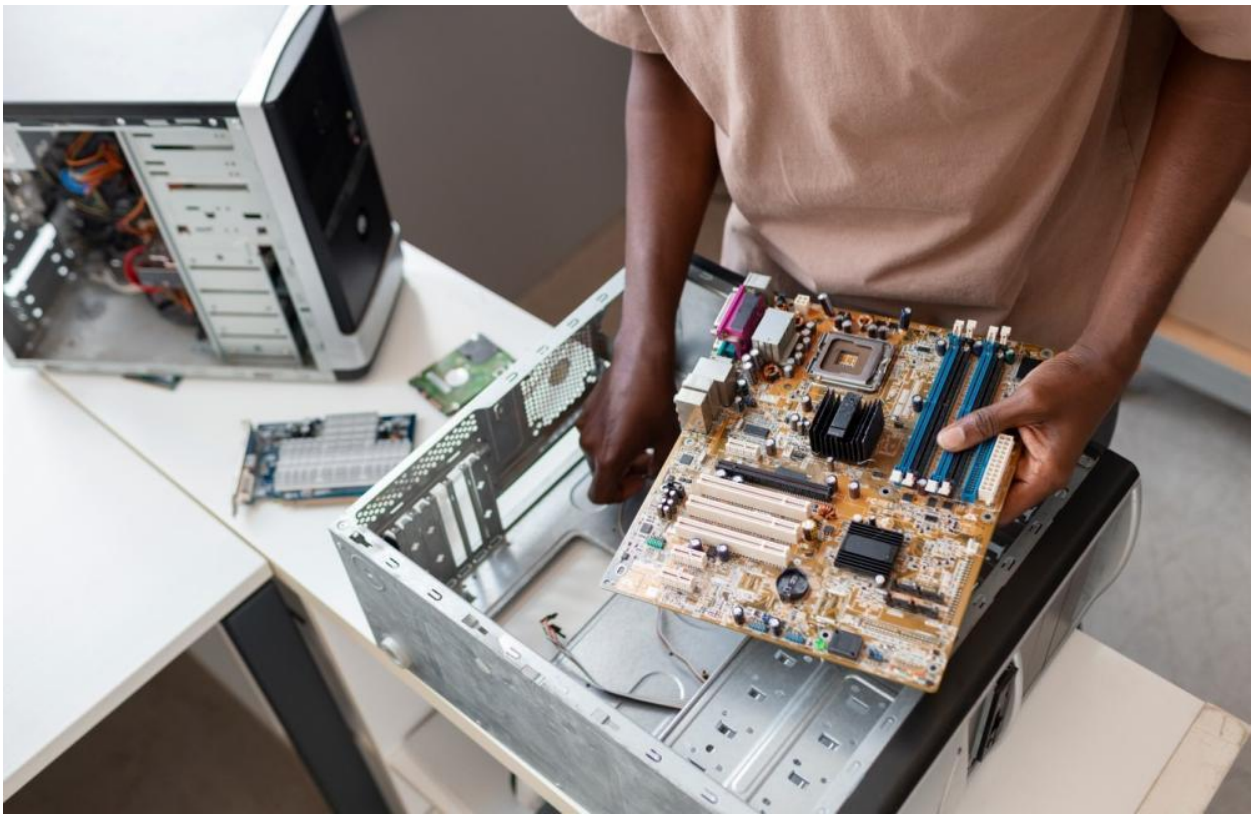
Caption: I disconnected the SATA data and power cables and removed the storage drive from its position in the cabinet.

Step 7: Remove the graphics card (GPU)



Caption: I removed the screws holding the graphics card and carefully released it from the PCIe slot.

Step 8: Remove the motherboard



Caption: After disconnecting the remaining cables, I removed the motherboard screws and carefully lifted the motherboard out of the cabinet.

Step 9: Final inspection



Caption: Finally, I arranged all the removed components separately and checked that all parts were safely removed.

3. **Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.**

Same as above

4. **Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.**

No.	Tool	Why is it important?
1	Screwdriver	Used to remove and tighten screws of the cabinet and components.
2	Anti-static wrist strap	Protects computer components from damage caused by static electricity.
3	Small container	Keeps screws and small parts safe and organized.
4	Thermal paste	Helps transfer heat from the CPU to the CPU cooler during assembly.
5	Cleaning brush	Removes dust from the motherboard, fans, and other components.

6	Compressed air can	Helps clean dust from hard-to-reach areas.
7	Flashlight	Helps see small screws, slots, and connections inside the cabinet.
8	Cable ties	Helps organize cables and maintain proper airflow inside the cabinet.

5. 2. Safety Precautions

No.	Safety precaution	Why is it important?
1	Switch off and unplug the PC.	Prevents electric shock and damage to components.
2	Wear an anti-static wrist strap.	Prevents static electricity from damaging sensitive components.
3	Work on a clean, dry surface.	Protects computer parts from dust, water, and accidental damage.
4	Handle components by their edges.	Prevents damage to delicate electronic parts and connectors.
5	Keep screws safely in a container.	Prevents losing screws or leaving them inside the cabinet.
6	Do not apply too much force.	Prevents damage to motherboard slots, cables, and connectors.
7	Keep liquids away from the PC.	Prevents short circuits and electrical damage.
8	Check cable connections before switching on the PC.	Ensures everything is connected correctly and safely.
9	Never open the PSU.	Its internal components can retain dangerous electrical charges even when unplugged.
10	Follow the motherboard manual.	Helps install components correctly and avoid compatibility problems.